# Star Vetter - confirmed eclipsing binary candidates

| TIC | Period (d) | Depth | BLS SNR | Pixel SNR | Offset | Tier |
|---|---|---|---|---|---|---|
| 437150434 | 5.2853 | 15.003% | 1081 | 85.33 | 1" | Clean |
| 237115819 | 14.7921 | 8.883% | 507 | 27.82 | 4" | Clean |
| 66732476 | 3.1087 | 9.814% | 370 | 26.93 | 8" | Clean |
| 29495157 | 5.1474 | 8.579% | 333 | 52.62 | 5" | Alias |
| 119459681 | 5.9442 | 15.977% | 311 | 68.15 | 2" | Clean |
| 411434067 | 1.3536 | 4.289% | 193 | 33.70 | 4" | Alias |
| 330901906 | 3.7321 | 0.498% | 60 | 87.24 | 4" | Thin margin |
| 303470667 | 5.4319 | 2.220% | 57 | 5.02 | 9" | Marginal |
| 273158289 | 1.1519 | 1.255% | 41 | 17.24 | 2" | Thin margin |
| 111149941 | 5.2401 | 0.045% | 10 | 9.73 | 21" | Ambiguous |
| 251898081 | 27.3089 | 18.630% | 5535 | 464.16 | 3" | Alias |
| 467214801 | 0.7609 | 0.950% | 2134 | 21.36 | 0" | Alias |
| 399770041 | 1.5802 | 4.291% | 489 | 36.92 | 1" | Clean |
| 349701853 | 30.3427 | 28.445% | 343 | 51.05 | 6" | Clean |
| 233459038 | 25.3384 | 8.315% | 237 | 84.46 | 5" | Alias |
| 281310348 | 1.7702 | 7.273% | 180 | 57.87 | 2" | Clean |
| 318887646 | 14.2817 | 3.605% | 150 | 6.94 | 13" | Clean |
| 444465823 | 18.5133 | 3.476% | 147 | 25.18 | 1" | Alias |
| 380943251 | 0.7209 | 7.667% | 101 | 15.26 | 3" | Clean |
| 66716482 | 8.0283 | 3.329% | 65 | 21.51 | 7" | Clean |
| 309656476 | 28.9204 | 0.050% | 25 | 13.80 | 14" | Clean |
| 229494833 | 15.3839 | 0.037% | 15 | 5.19 | 12" | Alias |
| 44533527 | 14.5952 | 0.101% | 12 | 7.33 | 16" | Clean |
| 220423668 | 3.6511 | 4.525% | 755 | 17.67 | 6" | Alias |
| 365998904 | 1.2494 | 2.398% | 375 | 15.52 | 3" | Clean |
| 233452636 | 1.9063 | 5.467% | 202 | 8.94 | 2" | Thin margin |
| 437171483 | 1.4735 | 4.947% | 923 | 73.37 | 3" | Clean |
| 411457490 | 2.4324 | 5.531% | 431 | 58.00 | 1" | Alias |
| 295373509 | 5.4219 | 8.026% | 247 | 22.12 | 2" | Clean |
| 374158907 | 14.6563 | 4.974% | 179 | 21.40 | 2" | Alias |

# TIC 333952894

Eclipsing binary candidate dossier - Star Vetter

**THIN MARGIN**

```
BLS EPHEMERIS
  Period        0.661146 d
  Epoch (t0)    1597.3614 BTJD
  Duration      0.1200 d
  Depth         8.703 %  (87,027 ppm)
  SNR           586.1

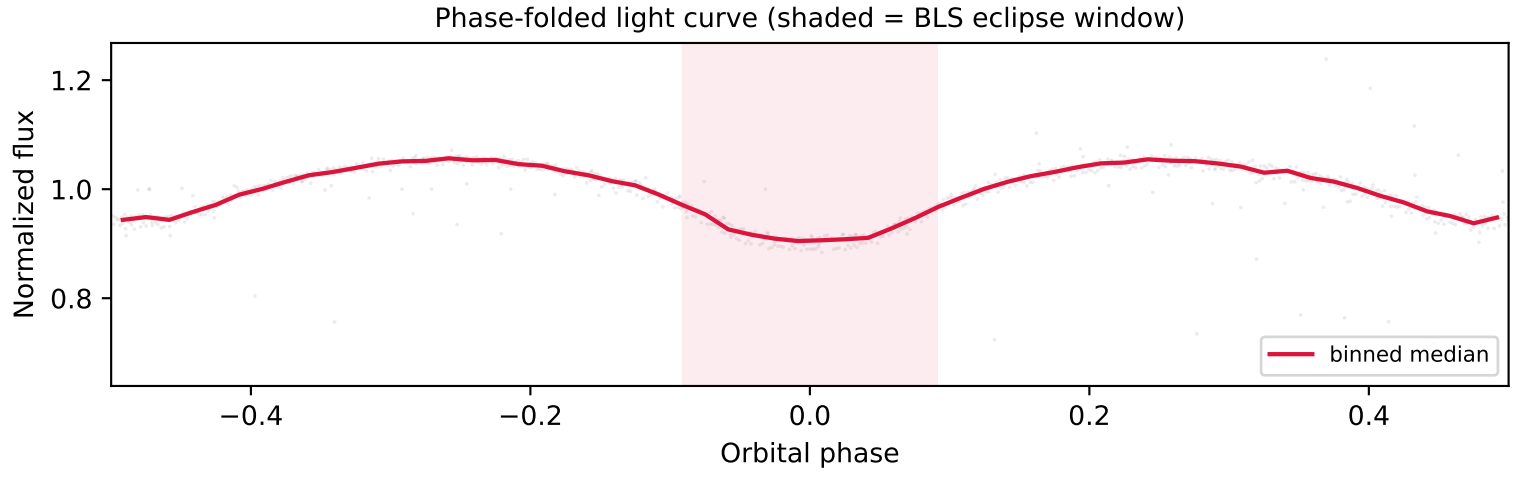
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      30  (gate >= 3)
  Robust depth ratio     1.18  (gate >= 0.50)
  Robust depth sigma     25.8  (gate >= 7.0)
  Odd/even z-score       0.31  (consistent)
  Secondary eclipse      not detected
```

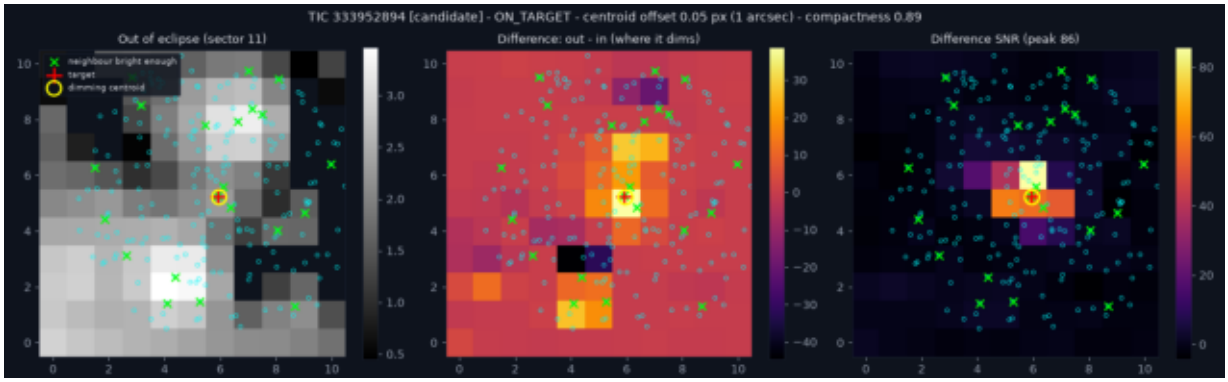
```
CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 11    light-curve dip in this sector: 84.8 sigma    t0 refined by -0.0100 phase
  Difference-image peak SNR   85.78   (gate >= 5.0)      Compactness   0.89   (gate >= 0.50)
  Dimming centroid offset     0.05 px (1 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 518050889
    0.07 px away, dTmag +1.18 - farther from the dimming than the target
```




Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.022 px (dimming centroid 0.050 px from target vs 0.071 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.

# TIC 378858431

Eclipsing binary candidate dossier - Star Vetter

**PERIOD ALIAS**

```
BLS EPHEMERIS
  Period       9.422447 d  (ALIASED, see below)
    -> corrected period: 18.844894 d
  Epoch (t0)   2891.0851 BTJD
  Duration     0.0840 d
  Depth        0.857 %  (8,570 ppm)
  SNR          291.8

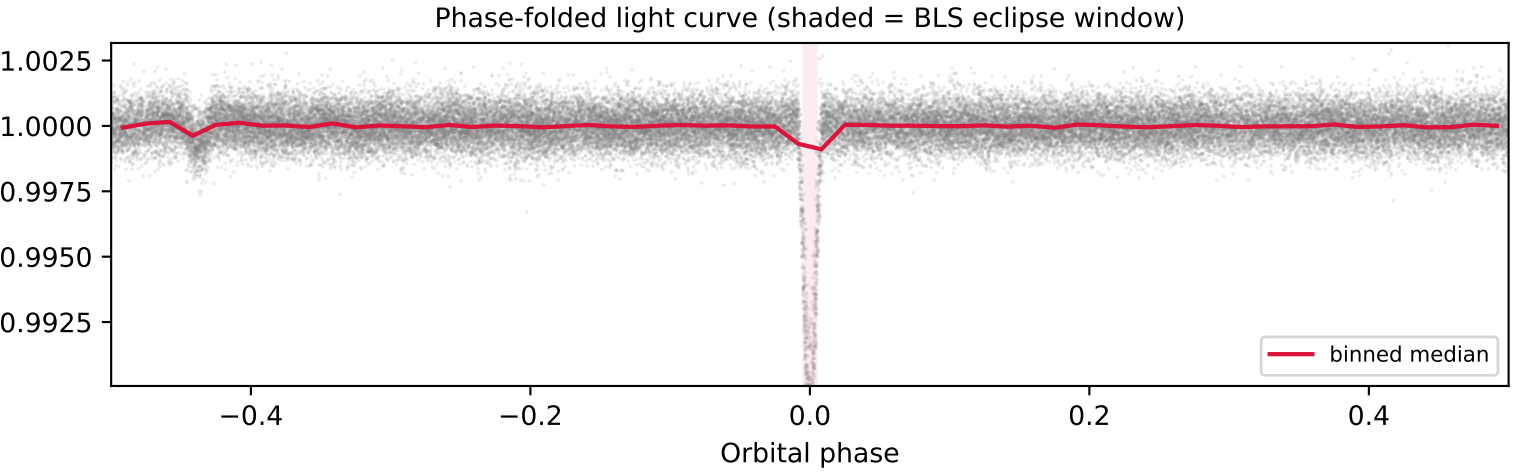
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     13  (gate >= 3)
  Robust depth ratio    1.03  (gate >= 0.50)
  Robust depth sigma    233.2  (gate >= 7.0)
  Odd/even z-score      4.24  (ALIAS FLAGGED)
  Secondary eclipse     detected, phase -0.43, 6.3 sigma
```

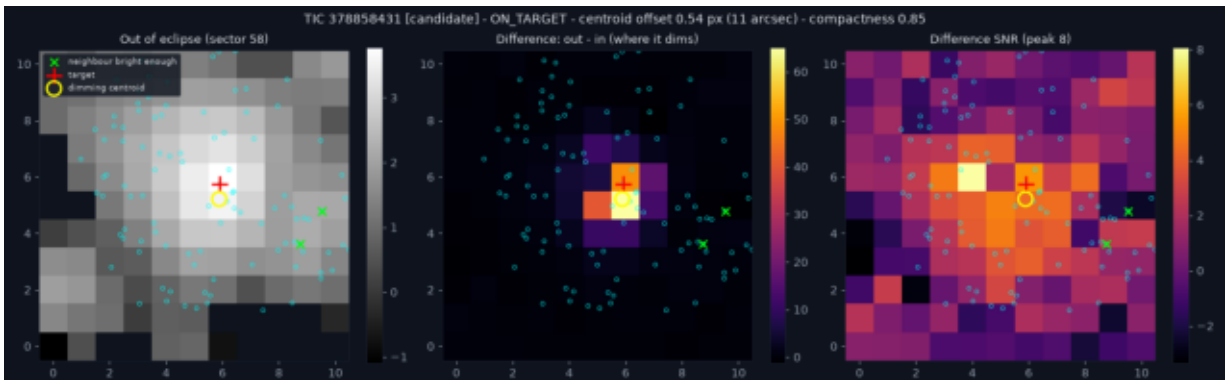
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 58    light-curve dip in this sector: 117.2 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   8.09   (gate >= 5.0)      Compactness   0.85   (gate >= 0.50)
  Dimming centroid offset     0.54 px (11 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 378858439
    3.28 px away, dTmag +4.71 - farther from the dimming than the target
```




Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 4.24 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 18.844894 d, in any write-up.